Proof By Induction

What is Proof by Induction?

Solution 2

Proof by induction establishes a statement for every integer from a starting value onwards.

Example: prove

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

for all positive integers n.

Checking $n = 3$:

$$1 + 2 + 3 = 6 = \frac{1}{2}(3)(4)$$

This confirms one case only. No finite list of examples proves the statement for all positive integers. We need a general argument.

How it works

Solution 3

Given $a_1 = 3$ and $a_{n+1} = a_n + 2$, prove that every term is odd.

Basis

The first term is 3, which is odd.

Inductive step

If a_k is odd, then $a_{k+1} = a_k + 2$ is also odd.

Conclusion

The first term is odd, so the second is odd, then the third, and so on. Every term is odd.

Think of a line of dominoes: push the first, and show that each falling domino knocks over the next.

Chapter Overview

Solution 4

1. **Summation proofs**

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

2. **Divisibility proofs**

Prove $n^3 - 7n + 9$ is divisible by 3.

3. **Matrix proofs**

$$\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^n = \begin{pmatrix} 1 & 1-2^n \\ 0 & 2^n \end{pmatrix}$$

Throughout this chapter, $n \in \mathbb{Z}^+$: the positive integers.

The Four Steps of Induction

Solution 5

1. **Basis**

Prove the statement for the starting value, usually $n = 1$.

2. **Assumption**

Assume it holds for an arbitrary positive integer $n = k$.

3. **Inductive step**

Use that assumption to prove the statement for $n = k + 1$.

4. **Conclusion**

State why the basis and inductive step together prove all positive integer cases.

Type 1: Summation Proofs

Solution 6

Prove

$$\sum_{r=1}^{n}(2r-1) = n^2$$

1. Basis

For $n = 1$, both sides equal 1.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k}(2r-1) = k^2$$

3. Induction

$$\sum_{r=1}^{k+1}(2r-1) = k^2 + [2(k+1)-1]$$

$$= k^2 + 2k + 1 = (k+1)^2$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

More on the 'Conclusion Step'

Solution 7

A complete conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$.
Therefore it holds for every positive integer n, by induction.

Make the logical chain explicit:

1. The statement is true at the starting value.
2. You assumed it for an arbitrary k.
3. You proved the next case from that assumption.
4. It therefore follows for all positive integers.

Equivalent clear wording is acceptable.

Further Example

Solution 8

Prove

$$\sum_{r=1}^{n} r2^r = 2[1 + (n-1)2^n]$$

1. Basis

At $n = 1, LHS = RHS = 2$.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r2^r = 2[1 + (k-1)2^k]$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + (k+1)2^{k+1}$$
$$= 2 + 2(k-1)2^k + (k+1)2^{k+1}$$
$$= 2 + 2k \cdot 2^{k+1}$$
$$= 2[1 + k2^{k+1}]$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$.
Therefore it holds for every positive integer n, by induction.

Test Your Understanding

Solution 9

Prove by induction

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

1. Basis

At $n = 1$, both sides equal 1.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + (k+1)^3$$

$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$

$$= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)]$$

$$= \frac{1}{4}(k+1)^2(k+2)^2$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8A - Core Pure 1 Pages 158-159

Solution 10

1. Prove

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

1. Basis

At $n = 1$, both sides equal 1.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r = \frac{1}{2}k(k+1)$$

3. Induction

$$\sum_{r=1}^{k+1} r = \frac{1}{2}k(k+1) + (k+1)$$

$$= \frac{1}{2}(k+1)(k+2)$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8A - Core Pure 1 Pages 158-159 — Solution 11

Prove by induction

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

1. Basis

At $n = 1$, both sides equal 1.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + (k+1)^3$$
$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$
$$= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)]$$
$$= \frac{1}{4}(k+1)^2(k+2)^2$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8A - Core Pure 1 Pages 158-159 — Solution 12

3a. Prove

$$\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n+1)(n-1)$$

1. Basis At $n = 1$, both sides equal 0.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r(r-1) = \frac{1}{3}k(k+1)(k-1)$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + k(k+1)$$
$$= \frac{1}{3}k(k+1)(k-1) + k(k+1)$$
$$= \frac{1}{3}k(k+1)(k+2)$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

3b. Hence find $\sum_{r=1}^{2n+1} r(r-1)$.

b

$$\frac{1}{3}(2n+1)(2n+2)(2n)$$
$$= \frac{4}{3}n(2n+1)(n+1)$$

Exercise 8A - Core Pure 1 Pages 158-159 — Solution 13

4a. Prove

$$\sum_{r=1}^{n} r(3r-1) = n^2(n+1)$$

1. Basis At $n = 1$, both sides equal 2.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$\sum_{r=1}^{k} r(3r-1) = k^2(k+1)$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + (k+1)(3k+2)$$
$$= (k+1)[k^2+3k+2]$$
$$= (k+1)^2(k+2)$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

4b. Find positive n if $\sum r^3 = 4\sum r(3r-1)$, both sums from 1 to n.

b

$$\frac{1}{4}n^2(n+1)^2 = 4n^2(n+1)$$
$$n+1 = 16 \Rightarrow n = 15$$

Exercise 8A - Core Pure 1 Pages 158-159 — Solution 14

5a. Prove

$$\sum_{r=1}^{n}(\frac{1}{2})^r = 1 - 2^{-n}$$

.

a

Basis: both sides are $\frac{1}{2}$. Assume the formula at k.

$$S_{k+1} = 1 - 2^{-k}$$
$$+2^{-(k+1)}$$
$$= 1 - 2^{-(k+1)}$$

5b. Prove

$$\sum_{r=1}^{n} r(r!) = (n+1)! - 1$$

.

b

Basis: both sides are 1. Assume the formula at k.

$$S_{k+1} = (k+1)! - 1$$
$$+(k+1)(k+1)!$$
$$= (k+2)! - 1$$

5c. Let

$$S_n = \sum_{r=1}^{n} \frac{4}{r(r+2)}.$$

Prove

$$S_n = \frac{n(3n+5)}{(n+1)(n+2)}.$$

c

Basis: both sides are $\frac{4}{3}$. Assume the formula at k.

$$S_{k+1} = \frac{k(3k+5)}{(k+1)(k+2)}$$
$$+\frac{4}{(k+1)(k+3)}$$
$$= \frac{(k+1)^2(3k+8)}{(k+1)(k+2)(k+3)}$$
$$= \frac{(k+1)(3k+8)}{(k+2)(k+3)}$$

Conclusion

Each proof establishes the basis and the step from k to $k+1$. Therefore each identity holds for every positive integer n, by induction.

6. A student claims

$$\left(\sum_{r=1}^{n} r\right)^2 = \sum_{r=1}^{n} r^2$$

They check $n = 1$, assume the statement at k, then simply write the statement at $k + 1$ and conclude it is true.

a. Identify the error.

b. Give a counterexample.

a

The inductive step has not been proved. Writing the next statement does not establish it from the assumption.

b

$n = 2: \quad (1 + 2)^2 = 9 \neq 1^2 + 2^2 = 5$

7. A student claims

$$\sum_{r=1}^{n} r = \frac{1}{2}(n^2 + n + 1)$$

They assume the formula at k, add $k + 1$ to both sides and correctly obtain the formula at $k + 1$, then conclude it is true.

a. Identify the error.

b. Give a counterexample.

a

The base case was never checked. A valid implication alone does not prove the starting statement.

b

$n = 1: \quad 1 \neq \frac{1}{2}(1 + 1 + 1) = \frac{3}{2}$

Exercise 8A - Core Pure 1 Pages 158-159 — Solution 17

Challenge: prove

$$\sum_{r=1}^{n}(-1)^r r^2 = \frac{1}{2}(-1)^n n(n+1)$$

1. Basis

At $n = 1$, both sides equal -1.

2. Assumption

Assume the statement holds for an arbitrary positive integer k:

$$S_k = \frac{1}{2}(-1)^k k(k+1)$$

3. Induction

Let S_k denote the sum in the assumption.

$$S_{k+1} = S_k + (-1)^{k+1}(k+1)^2$$
$$= (-1)^{k+1}(k+1)[-k/2 + k + 1]$$
$$= \frac{1}{2}(-1)^{k+1}(k+1)(k+2)$$

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Type 2: Divisibility Proofs — Solution 18

Prove $f(n) = 3^{2n} + 11$ is divisible by 4 for every positive integer n.

1. Basis

$f(1) = 9 + 11 = 20 = 4 \cdot 5$

2. Assumption

Assume $f(k) = 3^{2k} + 11$ is divisible by 4 for an arbitrary positive integer k.

Continue with the inductive step on the next slide.

Method 1: compare successive terms

Let $f(n) = 3^{2n} + 11$. The basis and assumption were established on the preceding slide.

3. Induction

$f(k+1) - f(k) = 9 \cdot 3^{2k} + 11 - (3^{2k} + 11)$

$= 8 \cdot 3^{2k} = 4(2 \cdot 3^{2k})$

Thus $f(k+1)$ is the sum of two multiples of 4.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Method 2: split off the assumed expression

With the same basis and assumption, rewrite $f(k+1)$ directly.

Induction and conclusion

$f(k+1) = 9 \cdot 3^{2k} + 11$

$= (3^{2k} + 11) + 8 \cdot 3^{2k}$

$= f(k) + 4(2 \cdot 3^{2k})$

Both terms are divisible by 4.

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Type 2: Divisibility Proofs

Solution 21

Prove $f(n) = 8^n - 3^n$ is divisible by 5 for every positive integer n.

1. Basis

$f(1) = 8 - 3 = 5$

This is divisible by 5.

2. Assumption

Assume $f(k)$ is divisible by 5 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 8 \cdot 8^k - 3 \cdot 3^k$

$= 3(8^k - 3^k) + 5 \cdot 8^k = 3f(k) + 5 \cdot 8^k$

Both terms are divisible by 5, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Test Your Understanding

Solution 22

Prove $f(n) = n^3 - 7n + 9$ is divisible by 3 for every positive integer n.

1. Basis

$f(1) = 1 - 7 + 9 = 3$

This is divisible by 3.

2. Assumption

Assume $f(k)$ is divisible by 3 for an arbitrary positive integer k.

3. Induction

$f(k+1) = k^3 + 3k^2 - 4k + 3$

$= f(k) + 3(k^2 + k - 2)$

Both terms are divisible by 3, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8B Core Pure 1 Page 162 — Solution 23

Let $f(n) = 13^n - 6^n$.

a. Show that $f(k+1) = 6f(k) + 7 \cdot 13^k$.

b. Prove by induction that $f(n)$ is divisible by 7 for every positive integer n.

1. Basis

$f(1) = 13 - 6 = 7$

This is divisible by 7.

2. Assumption

Assume $f(k)$ is divisible by 7 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 13 \cdot 13^k - 6 \cdot 6^k$

$= 6(13^k - 6^k) + 7 \cdot 13^k = 6f(k) + 7 \cdot 13^k$

Both terms are divisible by 7, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8B Core Pure 1 Page 162 — Solution 24

Let $f(n) = 5^{2n} - 6n + 8$.

a. Show that $f(k+1) = 25f(k) + 9(16k - 22)$.

b. Prove by induction that $f(n)$ is divisible by 9 for every positive integer n.

1. Basis

$f(1) = 25 - 6 + 8 = 27$

This is divisible by 9.

2. Assumption

Assume $f(k)$ is divisible by 9 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 25 \cdot 5^{2k} - 6k + 2$

$= 25(5^{2k} - 6k + 8) + 144k - 198$

$= 25f(k) + 9(16k - 22)$

Both terms are divisible by 9, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8B Core Pure 1 Page 162 — Solution 25

Prove $f(n) = 8^n - 3^n$ is divisible by 5 for every positive integer n.

1. Basis

$f(1) = 8 - 3 = 5$

This is divisible by 5.

2. Assumption

Assume $f(k)$ is divisible by 5 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 8 \cdot 8^k - 3 \cdot 3^k$

$= 3(8^k - 3^k) + 5 \cdot 8^k = 3f(k) + 5 \cdot 8^k$

Both terms are divisible by 5, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8B Core Pure 1 Page 162 — Solution 26

Prove $f(n) = 3^{2n+2} + 8n - 9$ is divisible by 8 for every positive integer n.

1. Basis

$f(1) = 81 + 8 - 9 = 80$

This is divisible by 8.

2. Assumption

Assume $f(k)$ is divisible by 8 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 9 \cdot 3^{2k+2} + 8k - 1$

$= f(k) + 8(3^{2k+2} + 1)$

Both terms are divisible by 8, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Exercise 8B Core Pure 1 Page 162 — Solution 27

Prove $f(n) = 2^{6n} + 3^{2n-2}$ is divisible by 5 for every positive integer n.

1. Basis

$f(1) = 64 + 1 = 65$

This is divisible by 5.

2. Assumption

Assume $f(k)$ is divisible by 5 for an arbitrary positive integer k.

3. Induction

$f(k+1) = 64 \cdot 2^{6k} + 9 \cdot 3^{2k-2}$

$= 64f(k) - 55 \cdot 3^{2k-2}$

Both terms are divisible by 5, so the next case holds.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Type 3: Matrix Proofs — Solution 28

Let

$$A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1 & 1-2^n \\ 0 & 2^n \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 1 & 1-2^k \\ 0 & 2^k \end{pmatrix}$$

3. Induction

$A^{k+1} = A^k A$

$$= \begin{pmatrix} 1 & 1-2^k \\ 0 & 2^k \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1-2^{k+1} \\ 0 & 2^{k+1} \end{pmatrix}$$

This is the proposed formula with $n = k + 1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.

Test Your Understanding — Solution 29

Let

$$A = \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1-3n & 9n \\ -n & 3n+1 \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 1-3k & 9k \\ -k & 3k+1 \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 1-3k & 9k \\ -k & 3k+1 \end{pmatrix} \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} -3k-2 & 9k+9 \\ -k-1 & 3k+4 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

Exercise 8C - Core Pure 1 Page 164 — Solution 30

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2(k+1) \\ 0 & 1 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

Exercise 8C - Core Pure 1 Page 164 — Solution 31

Let

$$A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 2k+3 & -4k-4 \\ k+1 & -2k-1 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

Exercise 8C - Core Pure 1 Page 164 — Solution 32

Let

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 2^k & 0 \\ 2^k - 1 & 1 \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 2^k & 0 \\ 2^k - 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 0 \\ 2^{k+1} - 1 & 1 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

Exercise 8C - Core Pure 1 Page 164 — Solution 33

Let

$$A = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 4k+1 & -8k \\ 2k & 1-4k \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 4k+1 & -8k \\ 2k & 1-4k \end{pmatrix} \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 4k+5 & -8k-8 \\ 2k+2 & -4k-3 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

b. Find n if

$$\begin{pmatrix} -1 & 3 \\ -2 & 5 \end{pmatrix} A^n = \begin{pmatrix} 11 & -21 \\ 10 & -19 \end{pmatrix}$$

b

$$\begin{pmatrix} 2n-1 & 3-4n \\ 2n-2 & 5-4n \end{pmatrix} = \begin{pmatrix} 11 & -21 \\ 10 & -19 \end{pmatrix}$$

$2n - 1 = 11 \Rightarrow n = 6$

Exercise 8C - Core Pure 1 Page 164 — Solution 34

Let

$$A = \begin{pmatrix} 2 & 5 \\ 0 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2^n & 5(2^n-1) \\ 0 & 1 \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 2^k & 5(2^k-1) \\ 0 & 1 \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 2^k & 5(2^k-1) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 5(2^{k+1}-1) \\ 0 & 1 \end{pmatrix}$$

This is the proposed formula with $n = k+1$.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$. Therefore it holds for every positive integer n, by induction.

b. Find $(A^n)^{-1}$.

b

$\det(A^n) = 2^n$

$$(A^n)^{-1} = \begin{pmatrix} 2^{-n} & 5(2^{-n}-1) \\ 0 & 1 \end{pmatrix}$$

Let

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 4 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 3^n & \frac{3^n-1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1-4^n}{3} & 4^n \end{pmatrix}$$

1. Basis

Substituting $n = 1$ into the proposed formula gives A itself, so the basis holds.

2. Assumption

Assume

$$A^k = \begin{pmatrix} 3^k & \frac{3^k-1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1-4^k}{3} & 4^k \end{pmatrix}$$

3. Induction

$$A^{k+1} = A^k A$$

The first and third diagonal entries become 3^{k+1} and 4^{k+1}. The off-diagonal entries are

$$3^k + \frac{3^k - 1}{2} = \frac{3^{k+1} - 1}{2}$$

$$\frac{1 - 4^k}{3} - 4^k = \frac{1 - 4^{k+1}}{3}$$

All other entries remain as required.

4. Conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$. Therefore it holds for every positive integer n, by induction.